Laboratory 2

Consider the problem from Laboratory 1 with the following modification:

A program is required to ensure the following postcondition:

> **Postcondition:** The initial matrix contains the filtered image.
>
> **Constraint: NO** other result matrix and **no** temporary matrix may be allocated! Only temporary vectors can be used, for which the space complexity fits within O(n) .
>
> **Note:** For the sequential implementation as well, it will be necessary to use temporary auxiliary vectors.
>
> **Objective:** Optimization of space complexity while achieving high performance.
>
> Input data is read from an input file named `"date.txt"` (the file must be created beforehand by adding randomly generated numbers).
>
> The result is saved in an `output.txt` file.

Implementation

a) Java

b) C++ (at least C++11)

Compared to Laboratory 1, only the following are required:

> Line-based distribution
>
> Kernel of size k = 3
>
> Dynamic allocation for C++

Testing: Measure the execution time for

> N = M = 10 and n = m = 3 ; p = 2 + sequential execution
>
> N = M = 1000 and n = m = 3 ; p = 2, 4, 8, 16 + sequential execution
>
> N = 10000 , M = 10000 and n = m = 3 ; p = 2, 4, 8, 16 + sequential execution

Notes:

Each test must be repeated 10 times, and the evaluation of the execution time is considered to be the arithmetic mean of the 10 runs.

For each execution (out of the 10) of each test case, use the same input file and verify correctness by comparing it with the output file obtained from the sequential execution.

Analysis:

Compare the performance for each case!

Compare the times obtained with the Java implementation versus the C++ implementation.

Evaluate the space complexity.

Deadline:

Week 5